Template Document

Academic Support Agent

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[IGS LaTeX 2ε guide]Journal of Glaciology authors' guide to the IGS LaTeX 2ε class file
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Abstract

The design for the *Journal of Glaciology* has been implemented as a LaTeX 2ε class file and is derived from article.cls. We recommend that authors use this guide as a template. While writing we suggest you use the two-column `[twocolumn]` option to check that mathematical equations fit the measure. Submitted papers must, however, be presented using the one-column `[review]` option. If you have any problems using the class file, please contact Overleaf support at `www.overleaf.com/contact`. The abstract should be less than 200 words and one paragraph long.

1 Using the IGS class file

Please ensure you have downloaded the latest version from http://igsoc.org/production/. The IGS LaTeX 2ε Journal guide has examples of most environments authors are likely to come across. The title page contains some new environments, e.g. affiliation and abstract. Papers should be divided into unnumbered sections with short section headings. SI units and internationally recognized systems of abbreviation should be used throughout. The TeX file should be named to reflect your paper number, i.e. 72A712.tex. Please remove any extraneous text (e.g. text from previous drafts, notes and comments that will not form part of the final printed text of the paper).

1.1 Typesetting the title page

In the IGS design, shortened versions of the title and authors are used in the running head. The shortened version is specified in square braces immediately after the `\title` and `\author` commands (see below).

```
\title[Short Title]{The Full Title of Your Paper}
\author[Short Authors]{Author 1, Author 2 and Author 3}
```

1.2 Lists

The IGS class file provides for numbered (`enumerate`) and unnumbered (`itemize`) lists. Nested lists are not encouraged. The default numbering system is 1., 2., 3., etc.; please do not change this unless there is a good reason. The IGS design removes bullet points from unnumbered lists.

1.3 User-defined macros

If possible, please do not define any new macros.

Table 1: One-column table captions will extend beyond the rules in two-column format. Do not try to adjust! Table captions do not have full points at the end

Period[a]	Surface elevation change	Emergence velocity
1975–85	−0.50	0.43
1986–2002	−1.03	0.32
Difference	−0.53	−0.11

[a]Please do not use more than one '&' between columns, and note that if a table includes table footnotes, it must be inside a `minipage` environment.

Table 2: Two-column table. Seasonal and annual SAT trends (°C decade^{-1}) in the Arctic

Area	1951–2005					1976–2005				
	Dec–Feb	Mar–May	Jun–Aug	Sep–Nov	Annual	Dec–Feb	Mar–May	Jun–Aug	Sep–Nov	Annual
Atlantic region	0.09	0.29	0.10	0.09	0.15	0.470	0.60	0.45	0.53	0.59
Siberian region	0.12	0.29	0.04	0.17	0.16	0.08	0.69	0.29	0.59	0.48
Pacific region	0.45	0.46	0.25	0.26	0.35	0.712	1.08	0.27	0.66	0.52
Canadian region	0.16	0.12	0.14	0.30	0.18	0.20	0.52	0.48	0.94	0.53
Baffin Bay region	-0.02	0.10	0.00	0.15	0.02	0.33	0.62	0.51	0.80	0.57
Arctic 1	0.16	0.21	0.12	0.20	0.18	0.36	200.65	0.42	0.74	0.54
Arctic 2	0.22	0.29	0.14	0.14	0.19	0.38	0.60	0.40	0.51	0.45
Arctic 3	0.28	0.31	0.14	0.13	0.21	0.42	40.53	0.41	0.42	0.43
NH (land + ocean)	0.13	0.13	0.10	0.10	0.12	0.27	0.24	0.25	0.25	0.25

1.4 Tables

Tables may be typeset in either one- or two-column format. To typeset two-column format, add asterisks (`\begin{table*}...\end{table*}`) as shown in Table **??**. We may change the format in-house if necessary. Please avoid the use of colour or shading. Note that if you choose to refer to tables using labels, `\caption` must precede `\label`, as in standard LaTeX. Vertical rules are not house-style and will be removed. Note the use of the minipage environment in Table **??** which enables table footnotes to be output. If the table is two-column, use `{178mm}` instead of `{86mm}` on line 6. The source code for Tables **??** and **??** is shown immediately below the tables.

1.5 Figures

Figures may be typeset in either one- or two-column format. One-column format allows up to 86 mm (e.g. Fig. **??**); two-column format up to 178 mm (e.g. Fig. **??**). Please do not provide original graphics files in which the figure is a great deal larger or smaller than what you envisage will be the final printed size. To typeset two-column format, add asterisks (`\begin{figure*}...\end{figure*}`) as shown in Fig. **??**. We may change the format in-house if necessary. Please note that if you choose to refer to figures using labels, `\caption` must precede `\label`, as in standard LaTeX.

Please send one file for each figure (in other words do not use subfigures) and use a name that clearly identifies it (e.g. '72A712Fig03.eps').

In addition, figures should be eps, ai (illustrator), ps, tif, psd or pdf. Use strong black lines with a width of at least 0.75pt at final printed size (avoid tinting if possible) and SI units in labels. Lettering should ideally be Optima to match the final typeface; Arial or a similar sans serif font for a second choice. Aim to have the final-size lettering at 9pt, if possible. Figures should not be in boxes. The source code for Figs **??** and **??** is shown immediately below the figures.

1.6 Equations

We are including some complex equations as examples. Equations should be checked for width by removing the `[review]` option. Note the use of `cases*` in the following equation:

$$\alpha_{t_2} = \begin{cases} \alpha_{t_1} - a_1[\ln(T+1)]\mathrm{e}^{(a_2\sqrt{n})} & n_\mathrm{d} > 0 \quad \text{and} \quad T > 0 \\ \alpha_{t_1} - a_3\mathrm{e}^{(a_2\sqrt{n})} & n_\mathrm{d} > 0 \quad \text{and} \quad T < 0 \\ \alpha_{t_1} + a_4 P_\mathrm{s} & n_\mathrm{d} = 0 \end{cases} \tag{1}$$

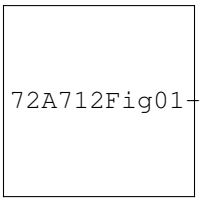


Figure 1: One-column figures should be $\leq$86 mm. Good artwork can make or break a paper. Capitalize the first word of a label and use round not square brackets for units.

Equations should be aligned on the equals signs where possible. Equations that extend beyond the one-column measure should be turned over before an operator.

$$l_c = l_0 \left(\frac{R_m}{R}\right)^2 \psi^{\frac{P}{P_0 \cos Z}} \times [\cos\beta\ \cos Z + \sin\beta\ \sin Z\ \cos(\psi_{\rm sun} - \psi_{\rm slope})] \tag{2}$$

1.7 Typesetting upright Greek characters

Normal greek: `$\alpha\beta\gamma\delta$` $\alpha\beta\gamma\delta$
Upright greek: `$\upalpha\upbeta\upgamma\updelta$`
Usual partial: `$\partial$` ∂
Upright partial: `$\uppartial$`

1.8 Marginal notes

The IGS class file redefines the LaTeX command `\marginpar`. If you wish to add a marginal note such as the one alongside this text, you would key `\marginpar{Editor! Help!}`. Marginal notes will be removed before printing.

Editor! Help!

1.9 References

All citations in text should include the author name(s) and the year of publication (e.g. 'Smith, 2014'; 'Smith and Jones, 2014'; 'Smith and others, 2015') and have an entry in the reference list.

References should:

- be short;
- be complete and accurate;
- be arranged in alphabetical order by first author's surname;
- include too much rather than too little information;
- include doi numbers where available (note that older bib databases often included doi's in the page field – in which case they may appear after a comma and without braces);
- include works accepted but not published as 'in press';
- not include personal communications, unpublished data or manuscript in preparation or submitted for publication, data published on the web (these should be included in the text).

1.9.1 Automatic references using BibTeX

To generate automatic references from a bib database, you must first specify the database (we are using `igsrefs.bib`) and then the IGS bibliography style by placing the following two commands where you would like the references to appear (normally at the end of your paper, before